Aeroacoustics Project 1

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December 4, 2025

Part A

A.1

The acoustic particle velocity u'_a of a time-harmonic plane wave can be estimated as $p' = \rho_0 c_0 u'_a$, where ρ_0 and c_0 are constants denoting the air density ($\rho_0 = 1.225$ kg/m^3) and speed of sound ($c_0 = 340$ m/s), we can calculate the root-mean-square value of the acoustic velocity $u_{a,rms}$ corresponding to a sound pressure level of $130\,dB$ with a reference pressure of $p_{ref} = 20\mu Pa$. Starting with the definition of sound pressure level:

$$SPL = 10log_{10}(\frac{p_{rms}^2}{p_{ref}^2}) \tag{1}$$

Thus, we can obtain p_{rms} as:

$$p_{rms} = \sqrt{p_{ref}^2 \cdot 10^{\frac{SPL}{10}}} = \sqrt{(20 \times 10^{-6})^2 \cdot 10^{13}} = 63.24\,Pa \tag{2}$$

Now, assuming that for this case $p_{rms} = p'$ we can calculate u'_a:

$$u'_a = \frac{p'}{\rho_0 c_0} = \frac{63.24}{1.225 \cdot 340} = 0.151\,m/s \tag{3}$$

A.2

The turbulence intensity Tu is a measure of the fluctuating velocity compared to the mean flow velocity U_∞, defined as $Tu = u_{rms}/U_\infty$. For a given turbulence intensity of $Tu = 5\%$ and a free stream velocity of $U_\infty = 30\,m/s$, the u_{rms} would be:

$$u_{rms} = Tu \cdot U_\infty = 0.05 \cdot 30 = 1.5\,m/s \tag{4}$$

We can see that the turbulence rates are larger by an order of 10 from the acoustic scales.

A.3

Having two independent, plane, harmonic, and stationary noise sources with pressure perturbations:

$$p'_j = \Re\left\{\hat{p}_j e^{i(\omega_j t+\phi_j)}\right\}, \quad j = 1, 2 \tag{5}$$

for the case of a simple harmonic wave, p_{rms} relates to the pressure amplitude such as:

$$p_{rms} = \frac{\hat{p}}{\sqrt{2}} \tag{6}$$

Assuming we have linear acoustics, we can add multiple independent noise sources:

$$p' = p'_1 + p'_2 \tag{7}$$

And the rms of the total pressure perturbation by definition

$$\begin{aligned} p_{rms}^2 &= \frac{1}{2T}\int_{-T}^{T} [p']^2\, dt = \frac{1}{2T}\int_{-T}^{T} (p'_1 + p'_2)^2\, dt \\ &= \frac{1}{2T}\int_{-T}^{T} \left([p'_1]^2 + 2\, p'_1 p'_2 + [p'_2]^2\right) dt \end{aligned} \tag{8}$$

Applying linearity:

$$\begin{aligned} p_{rms}^2 &= \frac{1}{2T}\int_{-T}^{T} [p'_1]^2 dt + \frac{1}{2T}\int_{-T}^{T} 2\, p'_1 p'_2 dt + \frac{1}{2T}\int_{-T}^{T} [p'_2(t)]^2 dt = \\ &= p_{1,rms}^2 + p_{2,rms}^2 + \frac{1}{2T}\int_{-T}^{T} 2\, p'_1 p'_2 dt \end{aligned} \tag{9}$$

Now, we can calculate the integral, expressing the perturbations fully in complex form using the complex conjugate:

$$\Re\{z\} = \frac{z + \bar{z}}{2} \tag{10}$$

Thus, for the pressure perturbations:

$$p'_j = \frac{1}{2}\hat{p}_j \left(e^{i(\omega_j t+\phi_j)} + e^{-i(\omega_j t+\phi_j)}\right) \tag{11}$$

Now for $p'_1 p'_2$ term

$$2\, p'_1 p'_2 = \frac{1}{2}\hat{p}_1\hat{p}_2 \left(e^{i(\omega_1 t+\phi_1)} + e^{-i(\omega_1 t+\phi_1)}\right)\left(e^{i(\omega_2 t+\phi_2)} + e^{-i(\omega_2 t+\phi_2)}\right) \tag{12}$$

Substituting into the integral:

$$\frac{1}{2T}\int_{-T}^{T} \frac{1}{2}\hat{p}_1\hat{p}_2 \left(e^{i[(\omega_1+\omega_2)t+\phi_1+\phi_2]} + e^{i[(\omega_1-\omega_2)t+\phi_1-\phi_2]} + e^{i[(\omega_2-\omega_1)t+\phi_2-\phi_1]} + e^{-i[(\omega_1+\omega_2)t+\phi_1+\phi_2]}\right) dt \tag{13}$$

The term consists of terms of the form e^{ix} and is bounded. We can take the limit as the number of samples becomes large, such as $T \to \infty$:

$$\begin{aligned}
\theta_1 &= (\omega_1 + \omega_2)t + (\phi_1 + \phi_2), \\
\theta_2 &= (\omega_1 - \omega_2)t + (\phi_1 - \phi_2), \\
\theta_3 &= (\omega_2 - \omega_1)t + (\phi_2 - \phi_1), \\
\theta_4 &= -(\omega_1 + \omega_2)t + (\phi_1 + \phi_2)
\end{aligned} \tag{14}$$

Now, taking the limit, we get

$$\lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} \frac{1}{2}\hat{p}_1\hat{p}_2 \sum_{k=1}^{4} e^{i\theta_k}\, dt = 0 \tag{15}$$

Thus:

$$p_{rms}^2 = p_{1,rms}^2 + p_{2,rms}^2 = \frac{1}{2}(\hat{p}_1^2 + \hat{p}_2^2) \tag{16}$$

A.4

Returning to equation (13) with $\omega_1 = \omega_2$:

$$\frac{1}{2T} \int_{-T}^{T} \frac{1}{2}\hat{p}_1\hat{p}_2 \left(e^{i[2\omega t+\phi_1+\phi_2]} + e^{i[\phi_1-\phi_2]} + e^{i[(\phi_2-\phi_1]} + e^{-i[2\omega t+\phi_1+\phi_2]}\right) dt \tag{17}$$

Now the term consists of both bounded and unbounded terms. As the bounded terms zero out, we'll look at the unbounded terms:

$$\begin{aligned}
&\frac{1}{2T} \int_{-T}^{T} \frac{1}{2}\hat{p}_1\hat{p}_2 \left(e^{i[\phi_1-\phi_2]} + e^{i[\phi_2-\phi_1]}\right) dt = \\
&= \Re(e^{i[\phi_1-\phi_2]})\hat{p}_1\hat{p}_2 \cdot \frac{1}{2T}\int_{-T}^{T} dt = \hat{p}_1\hat{p}_2\cos(\phi_1 - \phi_2)
\end{aligned} \tag{18}$$

equation (8) becomes:

$$p_{rms}^2 = p_{1,rms}^2 + p_{2,rms}^2 + \hat{p}_1\hat{p}_2\cos(\phi_1 - \phi_2) = \frac{1}{2}\left[\hat{p}_1^2 + \hat{p}_2^2 + 2\hat{p}_1\hat{p}_2\cos(\phi_1 - \phi_2)\right] \tag{19}$$

Looking at two noise sources that are in-phase, $\phi_1 = \phi_2$:

$$p_{rms}^2 = \frac{1}{2}(\hat{p}_1^2 + \hat{p}_2^2 + 2\hat{p}_1\hat{p}_2) = p_{1,rms}^2 + p_{2,rms}^2 + 2p_{1,rms}p_{2,rms} = (p_{1,rms} + p_{2,rms})^2 \tag{20}$$

A.5

The total SPL of the drone:

$$SPL_{tot} = 10\log_{10}\left(\frac{p_{rms,tot}^2}{p_{ref}^2}\right) \tag{21}$$

Since the acoustic sources are uncorrelated, we can use:

$$p_{rms,tot}^2 = \sum_{i=1}^{n} p_{i,rms}^2 = n\ p_{i,rms}^2 \tag{22}$$

And get:

$$SPL_{\text{tot}} = 10\log_{10}\left(\frac{n\,p_{i,\text{rms}}^2}{p_{\text{ref}}^2}\right) = 10\log_{10} n + 10\log_{10}\left(\frac{p_{i,\text{rms}}^2}{p_{\text{ref}}^2}\right) = 10\log_{10} n + SPL_i \tag{23}$$

A.6

Using the formula derived in the previous section, we can calculate the SPL level for a drone with n blades as n independent far-field acoustic sources.

$$SPL_i = 60\,dB\,(re\ p_{ref} = 20\ \mu Pa): \tag{24}$$

$$SPL_{\text{tot}}[dB(ref = 20\mu Pa), n = 4] = 66.02 \tag{25}$$

$$SPL_{\text{tot}}[dB(ref = 20\mu Pa), n = 6] = 67.78 \tag{26}$$

$$SPL_{\text{tot}}[dB(ref = 20\mu Pa), n = 8] = 69.03 \tag{27}$$

A.7

For correlated and in-phase rotors, the calculation of SPL_{tot} would be different, since $p_{tot,\text{rms}}$ is now combined differently, following the form given in equation (20):

$$p_{rms,tot}^2 = \left(\sum_{i=1}^{n} p_{i,rms}\right)^2 = n^2 p_{i,rms}^2 \tag{28}$$

And SPL_{tot} for this case would be:

$$SPL_{\text{tot}} = 10\log_{10}\left(\frac{n^2 p_{i,\text{rms}}^2}{p_{\text{ref}}^2}\right) = 10\log_{10} n^2 + 10\log_{10}\left(\frac{p_{i,\text{rms}}^2}{p_{\text{ref}}^2}\right) = 20\log_{10} n + SPL_i \tag{29}$$

$$SPL_{\text{tot}}[dB(ref = 20\mu Pa), n = 4] = 72.04 \tag{30}$$

$$SPL_{\text{tot}}[dB(ref = 20\mu Pa), n = 6] = 75.56 \tag{31}$$

$$SPL_{\text{tot}}[dB(ref = 20\mu Pa), n = 8] = 78.06 \tag{32}$$

A.8

Given a certain noise level of the drone SPL_1 measured at a distance r_1 from the drone, the overall attenuated sound pressure level of the drone at the observer location r_2 is SPL_2, and is computed by considering geometrical attenuation effect, as expressed below:

$$SPL_2 = SPL_1 - 10\log_{10}\left(\frac{r_2^2}{r_1^2}\right) \tag{33}$$

We can isolate r_2:

$$r_2 = r_1\sqrt{10^{0.1\cdot(SPL_1 - SPL_2)}} \tag{34}$$

Using the given data:

$$SPL_2 = 60\,dB\,(re\ p_{ref} = 20\ \mu Pa), \qquad r_1 = 1.5\ [m] \tag{35}$$

For the uncorrelated case:

$$r_2(n=4) = 3\,[m] \tag{36}$$

$$r_2(n=6) = 3.674\,[m] \tag{37}$$

$$r_2(n=8) = 4.242\,[m] \tag{38}$$

And for the correlated case:

$$r_2(n=4) = 6\,[m] \tag{39}$$

$$r_2(n=6) = 9\,[m] \tag{40}$$

$$r_2(n=8) = 12\,[m] \tag{41}$$

We can see that for the coherent rotors case, there is a larger distance for the SPL to decay, and its decay distance increases more with each additional rotor compared to the incoherent case.

Part B

B.1

Writing a Matlab script to calculate the lower, center, and upper bands of 1-octave, 1/3-octave, and 1/12-octave spectra. starting from the center frequency $f_c = 1000\,Hz$. We'll show the results in the tables below for 1 and 1/3-octave, the table for 1/12-octave is too large and can be seen in the code attached.

$Band$	fl_Hz	fc_Hz	fu_Hz
1	22.0971	31.25	44.1942
2	44.1942	62.5	88.3883
3	88.3883	125	176.777
4	176.777	250	353.553
5	353.553	500	707.107
6	707.107	1000	1414.21
7	1414.21	2000	2828.43
8	2828.43	4000	5656.85
9	5656.85	8000	11313.7

Table 1: 1-Octave Bands starting at $f_c = 1000$ Hz

B.2

Using $dB(A)$, we can see in Fig.1 that reducing the RPM of the propeller by half, thus reducing the frequency from 400 Hz to 200 Hz, will reduce the $dB(A)$ scaling by $\Delta dB(A) = -4.77 - (-10.85) \approx 6\, dB$. We can also see that increasing the RPM by a factor of 2 will increase the $dB(A)$ scaling by $\Delta dB(A) = -0.79 - (-4.77) \approx 4\, dB$.

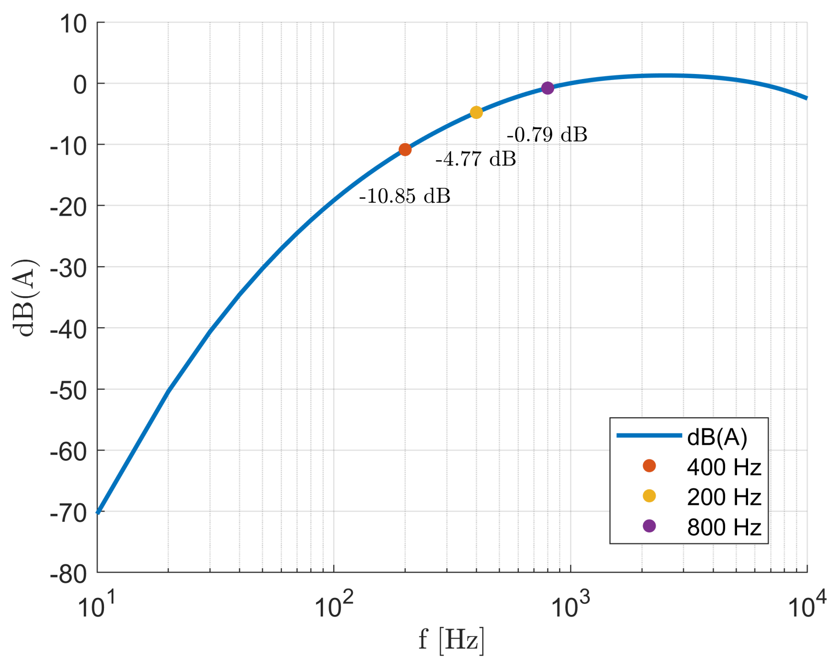


Figure 1: The dB(A) weighting scale, representing typical human ear sensitivity, with the effect of changes to the RPM of the propeller.

$Band$	fl_Hz	fc_Hz	fu_Hz
1	22.0971	24.8031	27.8406
2	27.8406	31.25	35.0769
3	35.0769	39.3725	44.1942
4	44.1942	49.6063	55.6812
5	55.6812	62.5	70.1539
6	70.1539	78.7451	88.3883
7	88.3883	99.2126	111.362
8	111.362	125	140.308
9	140.308	157.49	176.777
10	176.777	198.425	222.725
11	222.725	250	280.616
12	280.616	314.98	353.553
13	353.553	396.85	445.449
14	445.449	500	561.231
15	561.231	629.961	707.107
16	707.107	793.701	890.899
17	890.899	1000	1122.46
18	1122.46	1259.92	1414.21
19	1414.21	1587.4	1781.8
20	1781.8	2000	2244.92
21	2244.92	2519.84	2828.43
22	2828.43	3174.8	3563.59
23	3563.59	4000	4489.85
24	4489.85	5039.68	5656.85
25	5656.85	6349.6	7127.19
26	7127.19	8000	8979.7
27	8979.7	10079.4	11313.7
28	11313.7	12699.2	14254.4
29	14254.4	16000	17959.4

Table 2: 1/3 -Octave Bands starting at $f_c = 1000$ Hz

B.3

Given the empirical formula for SPL as a function of RPM:

$$SPL = C + 60log_{10}(RPM) \tag{42}$$

We can calculate ΔSPL:

$$\begin{aligned} \Delta SPL &= SPL(N \cdot RPM) - SPL(RPM) \\ &= 60\log_{10}(N \cdot RPM) - 60\log_{10}(RPM) \\ &= 60\log_{10}(N) \end{aligned} \tag{43}$$

Now, summing up the effects of the $dB(A)$ and SPL, we get that for increasing the RPM by a factor of 2, we have an increase in dB of around $22\,dB$, and reducing the RPM by half results in a reduction of $-24\,dB$.

Part C

C.1

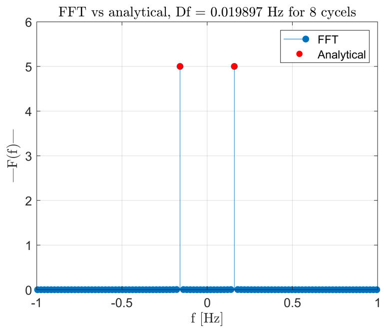
Our signal for this case is a simple sine wave:

$$f(x) = 10\sin(t) \tag{44}$$

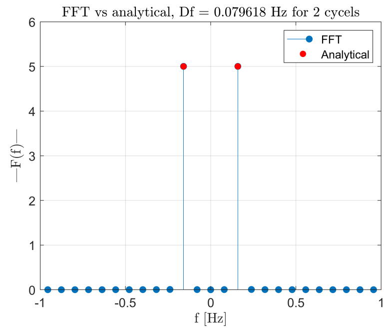
We can first calculate the Fourier transform analytically to be:

$$F(\omega) = -10\pi i(\delta(\omega - 1) - \delta(\omega + 1)) \tag{45}$$

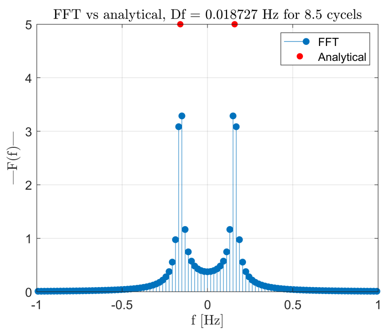
Resulting in a value of 0 to all frequencies except $\omega = \pm 1\,[rad/s]$. Using Matlab, we can calculate the Fourier transform with the `fft()` and use the inverse `ifft()` to obtain back the original signal. In the following plots, we'll show the comparison between the analytical and Matlab transformations, the signal vs. the `ifft()` output, and study the effects of low and high frequency resolution and number of cycles.



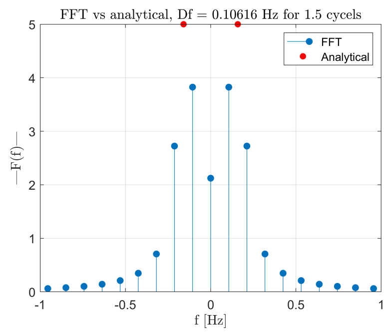
(a) High resolution, whole number of cycles case



(b) Low resolution, whole number of cycles case

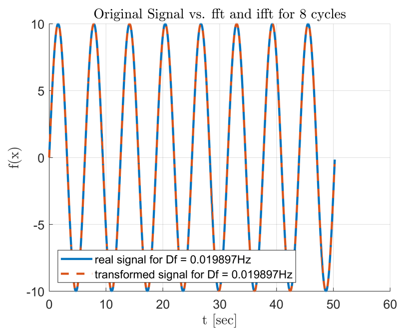


(c) High resolution, non-whole number of cycles case

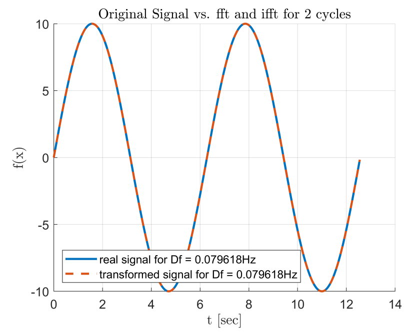


(d) Low resolution, non-whole number of cycles case

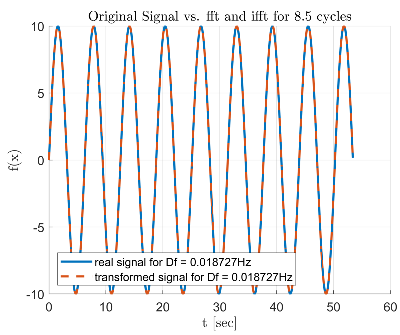
Figure 2: $F(\omega)$ for both analytical solution and `fft()` for different frequency resolutions.



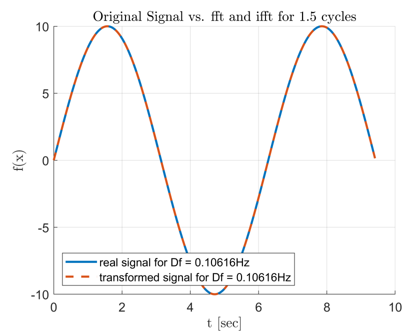
(a) High resolution, whole number of cycles case



(b) Low resolution, whole number of cycles case



(c) High resolution, non-whole number of cycles case



(d) Low resolution, non-whole number of cycles case

Figure 3: $f(x)$ for both analytical solution and `ifft()` for different frequency resolutions.

From Fig.2, we can see that for a whole number of cycles, the fft is identical to the analytical solution. For any non-whole number of cycles, we can see the `fft()` solution starting to differ from the analytical solution. In the differing solutions, we can clearly see the improvement of the `fft()` for higher resolutions. After taking the inverse `ifft()`, we can see that the numerical solution is nearly identical to the original signal.

C.2

To calculate the SPL from the microphone recording of the drone, we'll first calculate the PSD. The PSD was calculated numerically using two different algorithms:

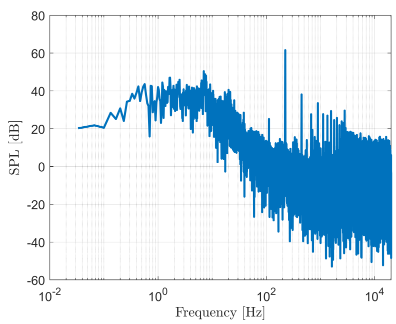
1. `fft()` - using Wiener–Khinchin theorem to calculate the S_{pp} on the transformed signal and from there obtaining $G_{pp} = 2S_{pp}$.

2. `pwelch()` using a 50% window overlap, a Hanning window and $\Delta f = 1.235\,Hz$

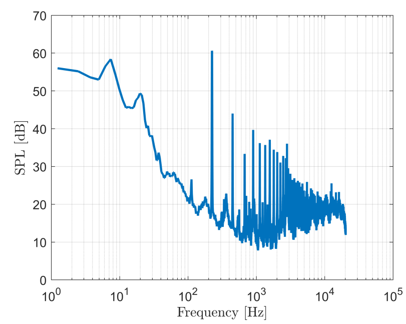
The SPL spectrum was calculated using the one-sided PSD, such as:

$$SPL = 10 \log_{10} \left(\frac{G_{pp} \Delta f}{p_{ref}^2} \right) \tag{46}$$

Plotting the results:



(a) `fft()` method.



(b) `pwelch()` method.

Figure 4: SPL spectrum of the noise signal measured by the microphone placed at $\theta = 105°$.

We can clearly see that the `pwelch()` method results in a cleaner spectrum, and the harmonic peaks are much more visible. Now, we'll check Parseval's theorem for both methods. From the code, we got a difference in the energy of $8E-5$ for `fft()` and $3E-3$ for `pwelch()`. Although the `pwelch()` method results in a cleaner spectrum, we have increased energy leakage by several orders of magnitude. Taking all that into consideration, we'll conclude that the `pwelch()` method is more ideal. Now, we'll plot the SPL spectrum with the frequency normalized by the blade passage frequency $f_b = \frac{\Omega}{60} B$ where B is the number of blades.

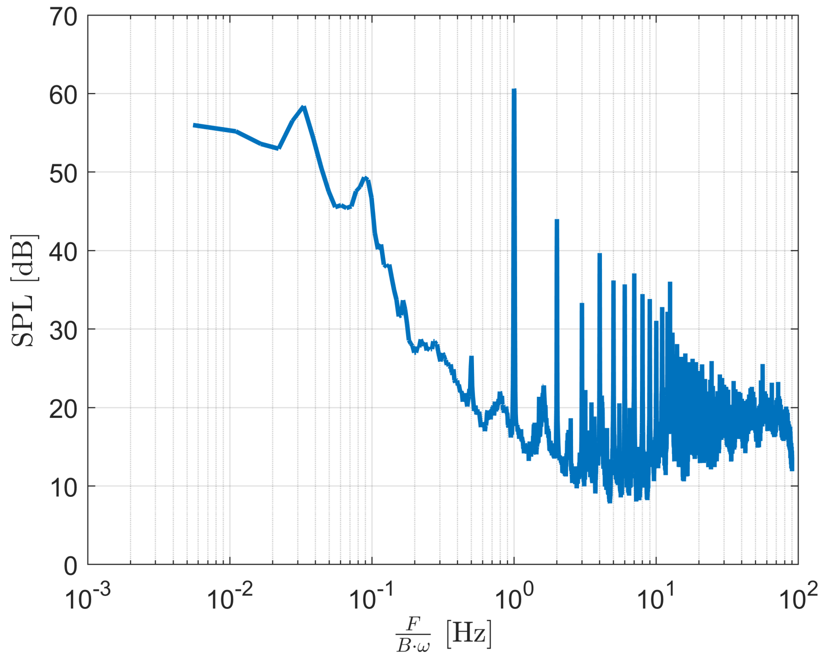


Figure 5: SPL spectrum of the noise signal measured by the microphone placed at $\theta = 105°$, normalized by the blade passage frequency.

We can see that the peaks in the spectra occur at each whole number of the normalized frequency. The first mode, which is the most energetic one, is at $f/f_b = 1$. From there on, each mode peaks with reducing energy as the frequency increases.

C.3

Computing the SPL spectra for all microphones using the `pwelch()` method, we can plot the SPL distribution with respect to θ:

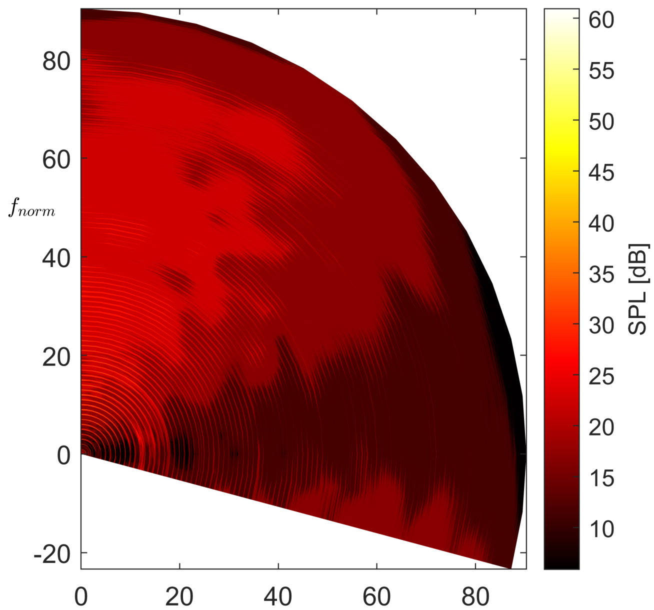


Figure 6: SPL spectrum of the noise signal over the entire microphone array.

We can see that above the drone, we get higher SPL numbers than to the side. This is because, as the drone produces a lift force upwards, air is sucked in towards the drone. The increased flow field above the drone produces hydrodynamic pressure perturbations that are higher than the acoustic perturbations. In fact, the microphones above the drones recorded more pseudo-sound, resulting in higher SPL values.